

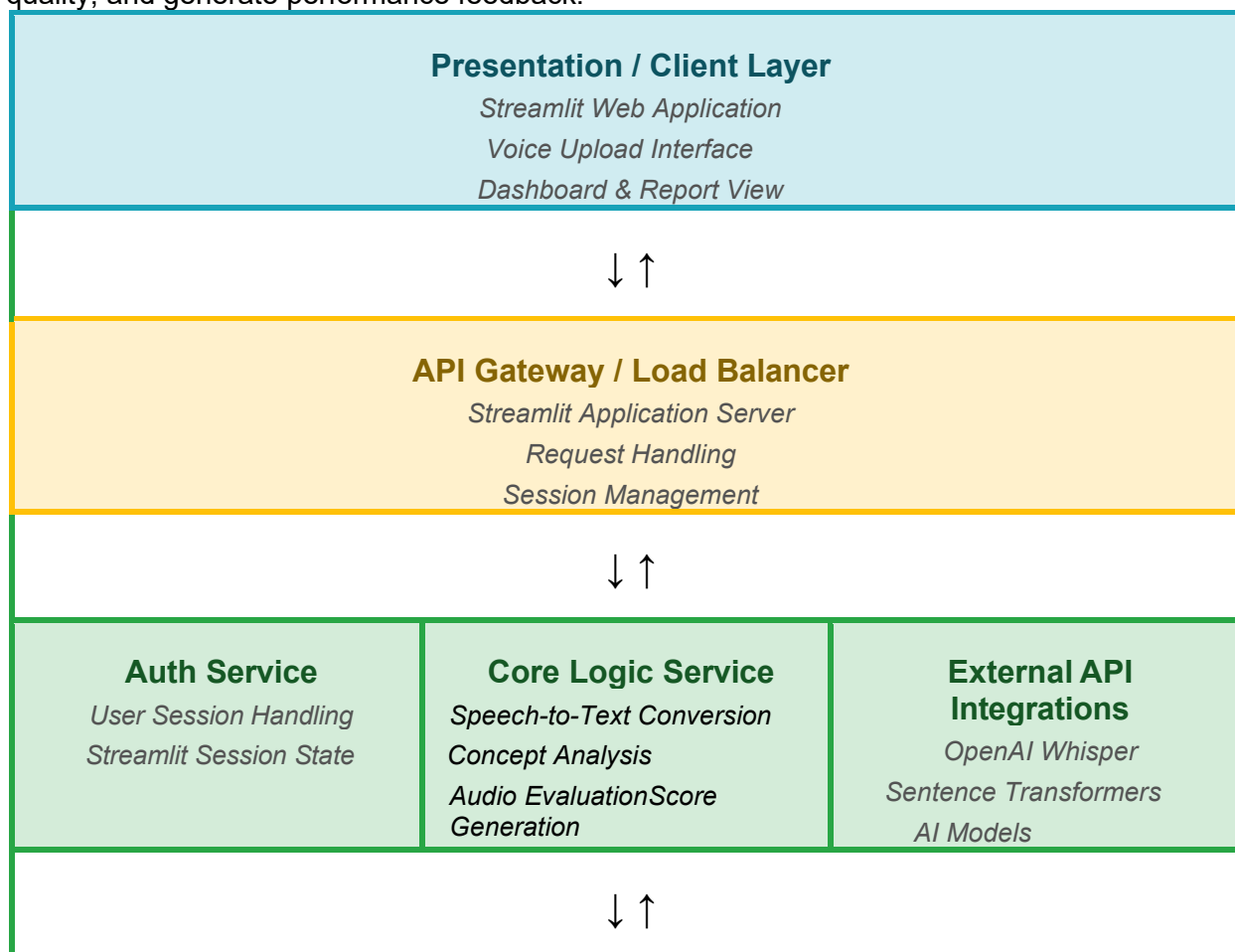
Solution Architecture

Date	04 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Solution Architecture Diagram:

The proposed architecture consists of a Streamlit-based user interface, AI processing modules, speech recognition system, NLP-based concept analysis engine, audio evaluation module, and report generation system.

Users provide voice input, which is processed using AI models to evaluate understanding, speech quality, and generate performance feedback.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides user interface for uploading voice, starting analysis, and viewing feedback reports</i>	<i>Streamlit</i>
<i>[API Gateway]</i>	<i>Handles user requests, application flow, and communication between UI and backend modules</i>	<i>Streamlit Server, Python</i>
<i>[Microservices]</i>	<i>Performs speech conversion, concept analysis, audio evaluation, scoring, and report generation</i>	<i>OpenAI Whisper, Sentence Transformers, Librosa, ReportLab</i>
<i>[Database]</i>	<i>Stores uploaded audio files, concept data, and generated analysis reports</i>	<i>Local File Storage, JSON Files</i>